

Mohammad Javad Zallaghi

PhD Candidate in Electrical Engineering: Robotics and Control

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Marital status: Single

[Google Scholar](#)



Education

- **PhD Candidate in Electrical Engineering** [2024 Sep – **NOW**]
[Vrije Universiteit Brussel \(VUB\)](#), Brussels, Belgium
Thesis: Learning-based Perception-Action Control for Provably Safe Aerial Robot Navigation
Supervisors: [Prof. Bram Vanderborght](#), [Prof. Adrian Munteanu](#), and [Dr. Bryan Convens](#)
- **MSc in Mechanical Engineering – Applied Mechanics** [2019 Sep – 2022 Mar]
Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran
Thesis: Design and Development of a Robotic ExoSuit for Upper Limb, **Thesis grade:** 20 out of 20
Supervisors: [Dr. Afshin Taghvaeipour](#), and [Dr. Hamed Ghafarirad](#)
- **BSc in Mechanical Engineering** [2015 Sep – 2019 Sep]
Khajeh Nasir Toosi University of Technology, Tehran, Iran
Thesis: Design and Fabrication of a Three-Wheel Robot with Performance Control, **Thesis grade:** 20 out of 20
Supervisors: [Prof. S. Ali A. Moosavian](#)

Research and Professional Roles

- **PhD Researcher in Robotics** [2024 – **NOW**]
Autonomous Navigation of Aerial Robots at [BruBotics](#), [R&MM](#), and [ETRO](#) Departments, Brussels, Belgium
- **Autonomous Vehicle and ADAS Algorithm Engineer** [2023 – 2024]
Decision, Planning, and Control Algorithm Engineer for Mobile Robots and ADAS
- **Research and Teaching Assistant** [2019-2022]
Amirkabir University of Technology, Multi-body Systems Lab and Intelligent Systems Research Centre
- **Research Assistant** [2018-2019]
Advanced Robotics & Automated Systems Lab (ARAS) in the K. N. Toosi University of Technology

Journal Papers

- **Zallaghi MJ**, Ghafarirad H, Taghvaeipour A. Direct model parameter identification of twisted string actuators using Nelder-Mead simplex method. *Proceedings of the Institution of Mechanical Engineers, Part C*. 2024;238(15):7747-7759. doi:[10.1177/09544062241233923](https://doi.org/10.1177/09544062241233923)

Selected Projects

- [SPEAR ROBOTICS](#) [EUROPEAN PROJECT]: Spatial Perception and Embodied Autonomy Research Project, contribute to Autonomous Navigation Policies for Robotics.

Skills and Courses

- **Robot software development:** C/C++, ROS/ROS 2, Gazebo, Webots, Python, PyTorch, Matlab/Simulink
- **Robot hardware development:** Nvidia Jetson Orin Computers, RealSense, ZED and OAK stereo cameras, 2D/3D LiDARs, Pixhawk autopilots, Holybro aerial robots, Dynamixel servos, Arduino and Raspberry Pi, CAD design with SolidWorks and Inventor
- **Reinforcement learning for robotics:** Nvidia Isaac Gym, Nvidia Isaac Sim, Aerial Gym, rl_games, Isaac Envs
- **Robotic courses:** Advanced Robotics, Reinforcement Learning, Dynamics of Multi-body System, Advanced Dynamics, Nonlinear Control, Adaptive Control, Artificial Neural Networks, Advanced Mathematics, Image Processing, Computational Intelligence and Fuzzy Systems

Honors

- IROS 2023 Humanoid Robot Competition: *MechaCheetahs*, **Gold Medal**, Champion of the competition, ranked 1st place: [\[Certificate\]](#)
- ICRA 2023 Humanoid Robot Competition: Reached the semifinals and placed within the top four: [\[Certificate\]](#)